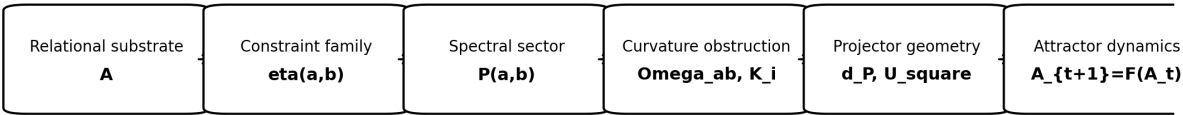


Relational Projector Geometry

A Pre-Geometric Correlation-to-Geometry Framework with Projector-Sector Attractor Dynamics

Research paper draft

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Relational Projector Geometry: correlation substrate -> projectors -> curvature -> geometry -> autonomous sector attractors

Figure 1. Conceptual pipeline of Relational Projector Geometry.

Abstract

Relational Projector Geometry is a pre-geometric framework in which geometry is reconstructed from spectral projectors defined over a relational correlation substrate. The basic object is a symmetric substrate A , interpreted as pre-geometric relational data rather than a metric on an already existing space. A constraint family $\eta(a,b)$ generates a family of operators whose spectral projectors $P(a,b)$ define the observable geometric sector. Projector variation produces a non-Abelian curvature obstruction Ω_{ab} , localized curvature profiles K_i , holonomy around small loops, and a projector distance d_P that functions as the primary metric-reconstruction observable. The framework also admits a preliminary autonomous dynamics: a self-consistent projector feedback update contracts trajectories toward gap-stable curvature-sector attractors. Numerical tests indicate that active sectors behave as correlation-supported relational defects, persist under perturbation, form internal correlation islands, and can coexist as weakly repulsive or co-stabilized multi-defect configurations. The present formulation does not claim to derive full spacetime field equations. Its strongest validated claim is narrower: relational projectors can reconstruct geometry-like observables and support preliminary autonomous attractor dynamics on a pre-geometric correlation substrate.

Keywords: pre-geometric physics; spectral projectors; emergent geometry; holonomy; curvature obstruction; correlation substrate; projector distance; attractor dynamics

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1. Motivation and Scope

The purpose of Relational Projector Geometry is to describe how geometry-like structure can emerge from relational information without assuming a background spatial manifold as fundamental. The framework begins with a substrate matrix A representing relational correlations. Geometry is not assigned directly to A . Instead, geometry is reconstructed through spectral projectors that respond to controlled relational constraints.

The current formulation should be read as a mathematical and computational framework for pre-geometric reconstruction, not as a completed physical theory. Its strongest validated components are kinematical: spectral projectors, projector distance, curvature obstruction, holonomy, and localized sector enrichment. Its dynamics component is preliminary but now numerically structured: autonomous feedback contracts projector trajectories toward stable curvature-sector attractors.

$$A \rightarrow P(a,b) \rightarrow \Omega_{ab} \rightarrow K_i \rightarrow S_r \rightarrow d_P$$

This chain is the operational core of the framework. The substrate A is not the emergent space. The projector family $P(a,b)$, its curvature response, and its distance geometry are the reconstructed observables.

2. Relational Substrate and Spectral Projectors

Let A be a real symmetric relational substrate. The entries of A are interpreted as pre-geometric relational correlations or couplings. A does not by itself define a physical metric. Instead, A is probed through a constraint family $\eta(a,b)$, where a and b are local deformation parameters.

$$H(a,b) = \eta(a,b) A \eta(a,b)$$

For a chosen spectral rank k , the top- k spectral subspace of $H(a,b)$ defines a projector $P(a,b)$. If $B(a,b)$ is an orthonormal basis of the selected subspace, then:

$$P(a,b) = B(a,b) B(a,b)^T$$

The projector P is the basic geometric observable. The framework does not identify individual substrate coordinates as spatial points. Instead, geometry is encoded in the way projector sectors vary under relational constraints.

2.1 Constraint-space locality

The parameters a and b define a local chart in constraint space. They are not necessarily physical coordinates. Their role is operational: they provide directions along which the projector sector can be differentiated, transported, and compared. This makes the framework closer to a projector-valued connection theory than to a conventional metric-first geometry.

3. Projector Curvature and Holonomy

Curvature enters through non-commutativity of projector response under two local constraint directions. Numerically, the derivatives are approximated by finite differences:

$$\partial_a P \sim [P(a+\delta, b) - P(a-\delta, b)] / (2\delta)$$

$$\partial_b P \sim [P(a, b+\delta) - P(a, b-\delta)] / (2\delta)$$

The local curvature obstruction is then defined by the projected commutator:

$$\Omega_{ab} = P [\partial_a P, \partial_b P] P$$

This object is the framework analogue of a non-Abelian curvature two-form. It measures whether the spectral sector responds integrably or non-integrably to local relational deformations.

3.1 Localized curvature profile

A node-level curvature profile is extracted from the row energy of the curvature operator:

$$K_i = \sqrt{\sum_j (\Omega_{ab})_{ij}^2}$$

The profile K_i identifies where curvature obstruction is concentrated in the substrate. It is later used to define active sectors and to construct autonomous feedback dynamics.

3.2 Holonomy around small loops

Given a small loop in constraint space, projectors at the loop vertices define transported bases and a loop holonomy U_{square} . For small loop area A_{square} , the logarithm of the holonomy approximates the curvature generator:

$$\log(U_{\text{square}}) / A_{\text{square}} \sim \text{projected curvature generator}$$

This is one of the strongest physical analogies in the framework. It connects Relational Projector Geometry to gauge-theoretic and Berry-curvature-like descriptions without assuming that the substrate already lives in a spacetime manifold.

4. Metric Reconstruction by Projector Distance

The primary metric-reconstruction observable is the Frobenius distance between projectors:

$$d_P(\alpha, \beta) = \| P_\alpha - P_\beta \|_F$$

This distance compares spectral sectors rather than substrate coordinates. When a family of constraints traces a trajectory or grid, d_P reconstructs the geometry of the projector response. This is the framework basis for treating emergent geometry as a geometry of distinguishable relational subspaces.

4.1 Why projector distance is central

- It is invariant under basis rotations inside the selected spectral subspace.
- It compares observable sectors rather than arbitrary coordinate labels.
- It naturally supports trajectory geometry, recurrence analysis, and attractor reconstruction.
- It is compatible with holonomy because both are defined from transported projector subspaces.

5. Curvature-Sector Defects and Core-Halo Structure

Active sectors are defined by ranking a sector score Q_i . The baseline score combines projector leverage and curvature activity:

$$Q_i = 0.65 * \text{normalized}(P_{ii}) + 0.35 * \text{normalized}(K_i)$$

For a sector size r , the active sector is:

$$S_r = \text{Top}_r(Q_i)$$

The core-halo enrichment diagnostic is:

$$C_r = [\sum_{i \in S_r} K_i / \sum_i K_i] / [r / N]$$

A core-halo obstruction is present when smaller sectors are more curvature-enriched than larger sectors:

$$C_4 > C_6 > C_8 > C_{12} > 1$$

This hierarchy means that curvature obstruction is not uniformly distributed. It concentrates in a small core and weakens into a surrounding halo. In the current interpretation, such sectors are localized relational curvature defects.

6. Autonomous Projector-Sector Dynamics

The autonomous dynamics studied in this version is a self-consistent projector feedback law. At each step the current projector and curvature profile define a reinforcement operator:

$$R_t = \text{Norm}(P_t D_{\{K,t\}} P_t)$$

The substrate is then updated by normalized relaxation toward this operator:

$$A_{t+1} = \text{Norm}((1-\lambda) A_t + \lambda R_t)$$

The update should not be interpreted as a final physical field equation. The numerical evidence supports a narrower statement: it is an autonomous projector-sector attractor rule. It contracts projector trajectories, preserves spectral gaps, stabilizes curvature sectors, and supports correlation-island defects.

6.1 Dynamics is not a simple scalar gradient flow

Several scalar variational interpretations were tested and rejected as global explanations. The update is not reliably described as descent of a simple alignment energy or ascent of an instantaneous alignment functional once the target R_t is recomputed at every step. The better interpretation is moving-target self-consistent relaxation:

$$A_t \rightarrow R[A_t], \text{ with } R[A_t] = \text{Norm}(P_t D_{\{K,t\}} P_t)$$

This distinction is important for the physics interpretation. The current dynamics is not yet an action principle. It is a validated attractor mechanism in projector geometry.

6.2 Robust parameter range

The most robust tested update strength is small, with the strongest basin-map result at λ approximately 0.02. Larger values can still stabilize sectors, but the cleanest projector contraction and recovery behavior appears in the smaller-update regime.

7. Correlation-Substrate Interpretation

A useful interpretation is to treat A as a pre-geometric correlation substrate. Define:

$$G_{ij} = |A_{ij}|$$

Active curvature sectors are then tested for internal correlation support. The correlation-island ratio compares average internal sector correlation to average boundary correlation:

$$\text{Island}(S) = \langle G_{ij} \rangle_{\{i,j \in S\}} / \langle G_{ij} \rangle_{\{i \in S, j \notin S\}}$$

The tests indicate that active sectors are correlation islands: they show stronger internal support than null and shuffled controls. However, they do not yet minimize a simple cut-entropy functional, so the correct statement is not that sectors are minimal entanglement cuts. The safer statement is that sectors are correlation-supported relational defects.

7.1 AdS/CFT-like analogy without literal identification

The framework can borrow one conceptual lesson from holographic reconstruction: geometry-like structure may be encoded in nonlocal correlation data rather than generated by local forces in an already existing space. Relational Projector Geometry does not claim an AdS bulk, a conformal boundary theory, or literal holographic duality. The analogy is methodological: pre-geometric correlations can support reconstructed geometry.

correlation substrate -> projector structure -> emergent metric/curvature observables

8. Numerical Validation Summary

The current validation suite supports the framework at three levels: static projector geometry, curvature-sector localization, and preliminary autonomous dynamics. The following tables summarize the main numerical claims used in this version.

Validation target	Feedback result	Null/control result	Interpretation
Attractor geometry	late dP step = $4.97\text{e-}15$; S4 persistence = 0.998	late dP step = 0.100; S4 persistence = 0.919	Feedback contracts projector trajectories toward stable sector states.
Attractor-basin robustness	best lambda = 0.02; late J = 0.610; late C4 = 1.783	Delta J over null = 0.137; Delta late step = -0.281	The attractor is robust across perturbation strength and update scale.
Claim robustness audit	persistence = 0.992; island = 1.459	persistence = 0.927; island = 1.259	Claims survive changes in sector definition and sector size.
Forced-proximity multi-defect test	repel = 0.675; co-stabilize = 0.223; merge = 0.000	repel = 0.620; co-stabilize = 0.156; merge = 0.086	Feedback favors separated co-stable defects over merging.

Table 1. Summary of principal autonomous-dynamics validation results.

Robustness claim	Pass rate or value	Status
Sector persistence gain across definitions	1.000	Strong
Correlation-island gain across definitions	1.000	Strong
Late projector-step reduction across definitions	1.000	Strong
Reference-distance reduction across definitions	0.667	Moderate
Sector-recovery gain across definitions	0.711	Moderate
Enrichment gain across definitions	0.556	Weak-to-moderate

Table 2. Final claim-robustness audit across sector definitions, sector sizes, and perturbations.

8.1 Figures from validation tests

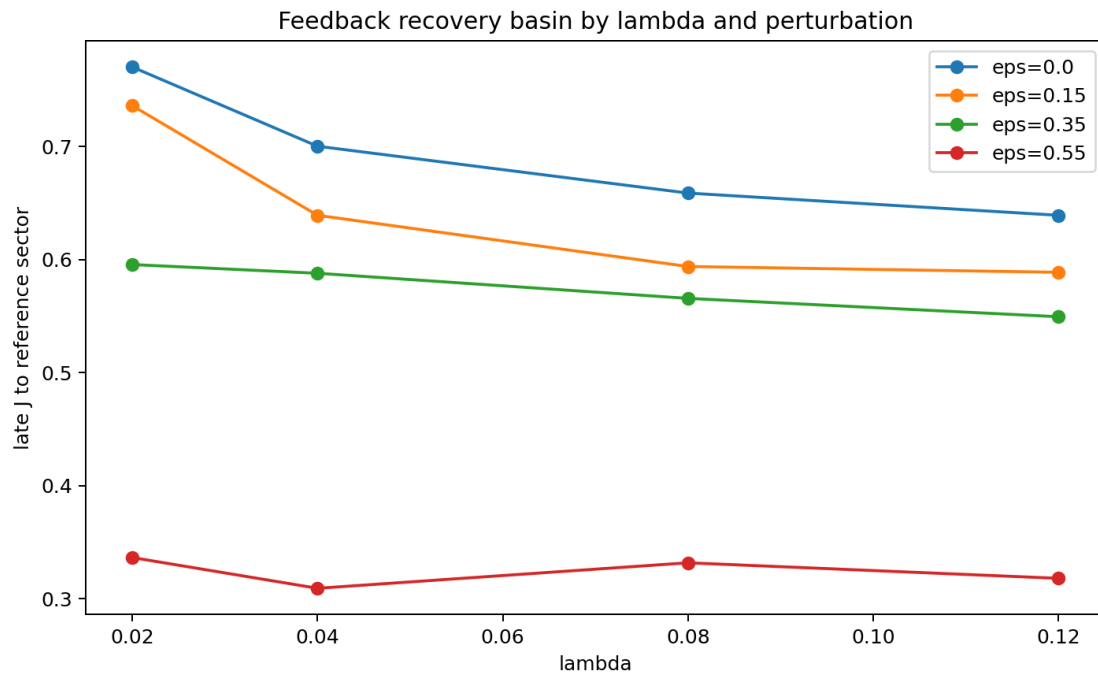


Figure 2. Recovery basin by feedback strength and perturbation amplitude.

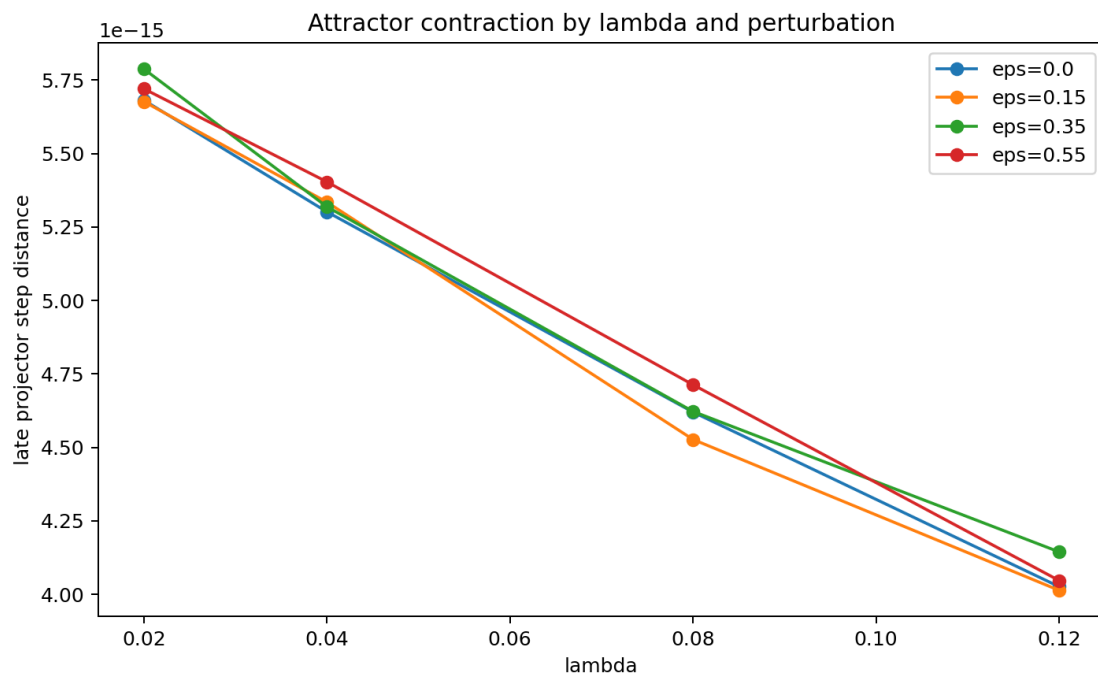


Figure 3. Projector-step contraction across the attractor basin.

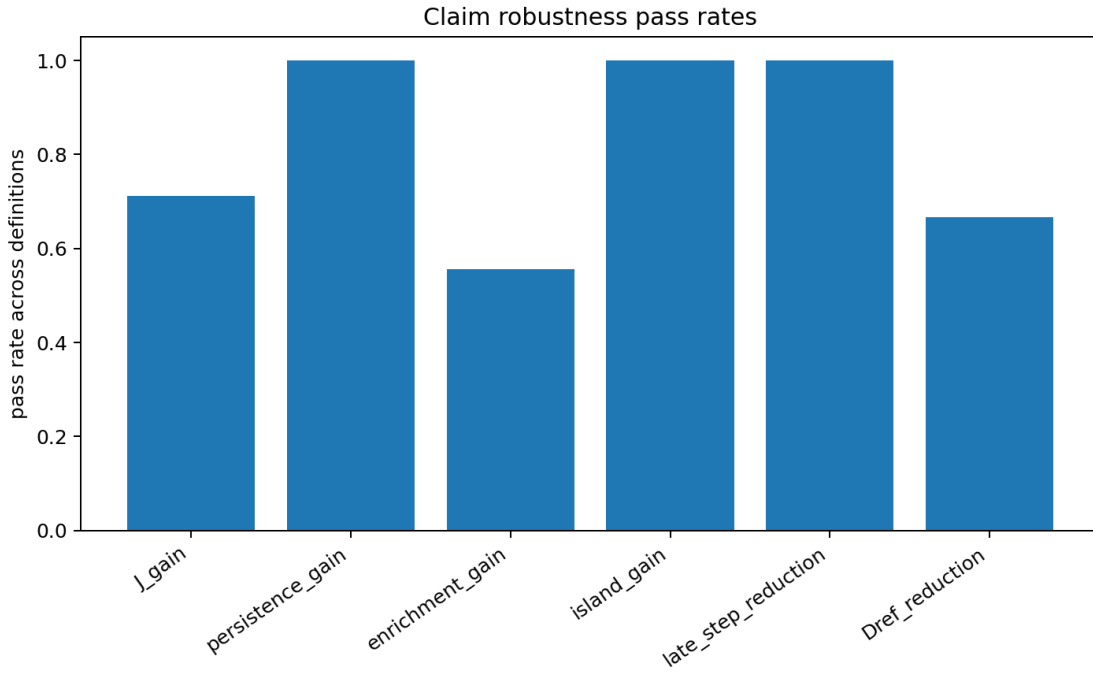


Figure 4. Claim-robustness pass rates across sector definitions and sector sizes.

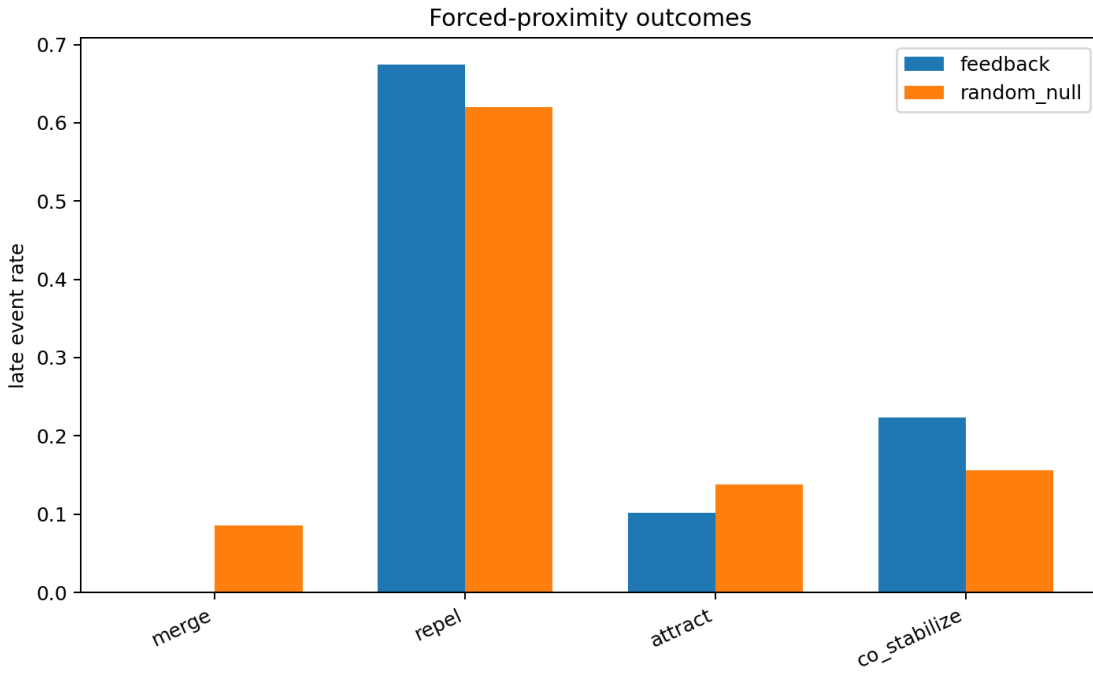


Figure 5. Forced-proximity multi-defect interaction outcomes.

9. Physical Interpretation and Limitations

The present framework is most naturally interpreted as a pre-geometric projector theory. It does not assume that local interactions occur inside a pre-existing space. Instead, it reconstructs geometry-like observables from relational correlations and spectral sector response.

9.1 What is currently supported

- Projector distance d_P functions as a metric-reconstruction observable over constraint-induced spectral sectors.
- Projected commutators define a curvature obstruction Ω_{ab} .
- Holonomy around small constraint loops behaves like a curvature diagnostic.
- Curvature profiles K_i define localized core-halo sectors.
- Autonomous feedback produces gap-stable projector-sector attractors.
- Active sectors behave as correlation-supported relational defects.
- Multiple sectors can coexist and show weak separation/co-stabilization behavior.

9.2 What is not yet supported

- No derivation of Einstein equations is claimed.
- No Lorentzian causal structure has been derived yet.
- No stress-energy analogue is currently defined.
- The autonomous update is not yet a known action principle or scalar gradient flow.
- Multi-defect behavior is weakly interaction-like but not yet particle-like dynamics.
- The AdS/CFT analogy is conceptual only; no literal holographic duality is asserted.

10. Development Roadmap

The next stage should not simply add more numerical tests of the same kind. The framework now needs theoretical compression: a smaller set of definitions, a clearer hierarchy of claims, and a more physical dynamics principle. The following directions are recommended.

10.1 Variational or action-based dynamics

The major open theoretical target is a principled dynamics. The current feedback rule is empirically useful but not yet derived from an action. A future action candidate may involve holonomy curvature, sector enrichment, projector-distance contraction, or correlation-island stability.

$$S_{\text{RPG}} = \text{sum_squares}(\log U_{\text{square}}) + \text{sector/correlation regularization}$$

10.2 Lorentzian and causal extension

A physically stronger version must define causal structure. This could require signed or indefinite relational operators, hyperbolic constraint families, or a distinction between time-like and space-like projector response.

10.3 Multi-defect dynamics

The current multi-defect result supports coexistence, persistence, weak repulsion, and co-stabilization. The next stage should test controlled defect scattering, defect annihilation, defect creation thresholds, and whether a conserved or approximately conserved quantity governs sector interactions.

10.4 Correlation-reconstruction program

The correlation-substrate interpretation should be expanded. The framework should test whether d_P , curvature, and sector structure can be predicted from correlation observables alone, and whether a boundary-correlation-to-bulk-geometry reconstruction map can be formalized without invoking literal AdS/CFT.

10.5 Benchmark standardization

Future work should package the numerical tests as a reproducible benchmark suite with fixed seeds, parameter sweeps, null controls, and clear pass/fail criteria. This will make the framework easier to evaluate independently.

Appendix A. Core Definitions

Relational substrate: A is a symmetric pre-geometric correlation/coupling matrix.

Constraint operator: $\eta(a,b)$ defines local relational deformations used to probe A .

Constrained operator: $H(a,b) = \eta(a,b) A \eta(a,b)$.

Projector: $P(a,b) = B(a,b)B(a,b)^T$ for a selected rank- k spectral sector.

Projector distance: $d_P(\alpha, \beta) = \|P_\alpha - P_\beta\|_F$.

Curvature obstruction: $\Omega_{ab} = P[\partial_a P, \partial_b P]P$.

Curvature profile: $K_i = \sqrt{\sum_j \Omega_{ij}^2}$.

Active sector: $S_r = \text{Top}_r(Q_i)$, with Q_i based on P_{ii} , K_i , and optionally correlation strength.

Core-halo enrichment: $C_r = [\sum_{i \in S_r} K_i / \sum_i K_i] / [r/N]$.

Feedback operator: $R_t = \text{Norm}(P_t D_{\{K,t\}} P_t)$.

Autonomous update: $A_{t+1} = \text{Norm}((1-\lambda)A_t + \lambda R_t)$.

Correlation substrate: $G_{ij} = |A_{ij}|$.

Correlation island: A sector whose internal correlation exceeds its boundary correlation.

Appendix B. Recommended Paper-Safe Claims

- Relational Projector Geometry reconstructs geometry-like observables from spectral projectors over a relational substrate.
- The projected commutator Ω_{ab} defines a local curvature obstruction in constraint space.
- Projector distance d_P is the main metric-reconstruction observable.
- Curvature obstruction localizes into core-halo sectors measured by C_r .
- Autonomous feedback contracts projector trajectories toward gap-stable curvature-sector attractors.
- Active sectors behave as correlation-supported relational defects.
- The current dynamics is preliminary and should not be presented as a complete spacetime field equation.